

HealthConnect Week 8 Final Analytics & Decision Support Report

Final Data Analytics Package

1. Executive Summary

The HealthConnect Data Analytics work progressed from data preparation and exploratory analysis to deeper segmentation, validation and decision support. The final dashboard analyses 5,000 appointments and identifies important differences in no-show behaviour across previous no-show history, booking lead time, distance to clinic, reminder channel, appointment type, gender, appointment day and risk group.

The validated core outcome is a 48.46% no-show rate (2,423 of 5,000 appointments), compared with a 46.28% attendance rate (2,314) and a 5.26% cancellation rate (263). Previous no-show history, longer booking lead times, greater distance and the high-risk segment show materially different observed no-show rates and therefore provide candidate areas for targeted operational investigation and predictive modelling.

The results are observational. They should support prioritisation and hypothesis generation rather than be interpreted as proof that any single factor causes a patient to miss an appointment.

2. Key Visualisations and Results

Visualisation	Result	Decision-support meaning
Overall appointment outcome	No-show 48.46%; attended 46.28%; cancelled 5.26%	Establishes the overall appointment utilisation problem.
Previous no-show behaviour	Yes 55.41%; No 43.51%	Previous behaviour is a useful candidate risk indicator.
Booking lead group	27.81% → 33.55% → 43.21% → 60.49%	Longer booking lead groups show higher observed no-show rates.
Distance to clinic	Above 30 km ≈ 68.1%	Long-distance appointments warrant further operational investigation.
Reminder channel	No reminder 51.39%; SMS 45.75%; Email 48.41%; WhatsApp 49.77%	Reminder groups differ, but causal effectiveness is not established.
Appointment type	Follow-up 51.23%; Diagnostic 49.75%; Specialist 47.44%; General 46.64%	Appointment context is another useful segmentation dimension.
Risk score group	High 69.98%; Medium 55.70%; Low 38.55%	Risk segmentation separates groups with materially different observed outcomes.
Appointment day	Sunday 50.47% is highest in the displayed day comparison	Day-of-week patterns can be investigated operationally.

3. Updated Business Insights

- **High overall no-show burden:** Nearly half of all appointments were recorded as no-shows (48.46%), making attendance management a central operational issue in the dataset.
- **Previous behaviour matters:** The previous-no-show group has a 55.41% observed no-show rate versus 43.51% without previous no-show history.
- **Longer booking lead is associated with higher no-show rates:** The rate rises from 27.81% for 0–7 days to 60.49% for 31–60 days.
- **Distance is a high-priority segment:** The above-30 km group records approximately 68.1% no-show rate in the dashboard.
- **Risk segmentation has practical value:** High-risk appointments show 69.98% no-show rate versus 55.70% medium and 38.55% low.
- **Reminder channels require adjusted testing:** No Reminder is 51.39%, SMS 45.75%, Email 48.41% and WhatsApp 49.77%; however, channel assignment may not be random.

4. Updated Recommendations

1. Create a risk-based appointment monitoring workflow that uses validated risk indicators and is tested by Data Science before operational deployment.
2. Prioritise patients/appointments with previous no-show history for further investigation and appropriate confirmation workflows.
3. Investigate long-lead appointments and evaluate whether confirmation closer to the appointment date reduces missed appointments.
4. Investigate barriers associated with appointments above 30 km, including travel burden and service accessibility.
5. Run controlled or adjusted analyses of reminder channels before selecting a channel as an intervention.
6. Use appointment type and appointment-day patterns to inform operational scheduling analysis.
7. Build a feedback loop in which Data Science model findings are compared against the descriptive dashboard and unexpected differences are investigated.

5. Analytics Report / Methodology

The analytical workflow used cleaned appointment data, derived outcome flags, grouped booking lead time and distance variables, KPI calculations, pivot-style validation, segmentation and Power BI decision-support visualisations.

- Count appointments and outcomes using COUNT/COUNTIFS or Power BI COUNTROWS/CALCULATE.
- Calculate rates as the relevant outcome count divided by the relevant appointment population.
- Use grouped dimensions for booking lead time, distance, age and risk score.
- Compare no-show rates across previous no-show, reminder channel, appointment type and other segments.
- Validate Power BI KPI values against independent Excel calculations.
- Use risk groups to compare observed no-show rates rather than relying only on aggregate appointment counts.
- Document the denominator for every executive KPI.

6. Analytical Limitations

- The analysis is observational; it identifies associations rather than causal effects.
- Reminder-channel comparisons may be confounded by differences in patient and appointment characteristics.
- Small segment sizes can make percentages unstable; counts should be shown alongside rates where possible.

- Missing-value treatment may influence grouped results.
- The dataset may not include all operational, socioeconomic or clinical reasons for non-attendance.
- The dataset is not sufficient by itself to generalise the findings to all healthcare populations.
- The dashboard KPI labelled 'Reminder Effectiveness 4.03%' requires a documented definition before it should be treated as a final executive KPI.
- The 'Reminder Rate 72.68%' also requires a documented numerator and denominator.
- Risk-score charts displaying values such as 3.0K, 2.1K and 6.2K appear to be aggregated risk-score sums rather than appointment counts; labels should make this explicit or be replaced by count/rate visuals.

7. Cross-Track Contribution: Data Science

The Data Analytics track should provide Data Science with the validated descriptive findings and KPI definitions so that predictive modelling can test whether the observed patterns remain useful after considering multiple variables.

Information supplied to Data Science

- No-show rate = 48.46% and outcome counts.
- Previous no-show: 55.41% versus 43.51%.
- Booking lead groups: 27.81%, 33.55%, 43.21%, 60.49%.
- Distance above 30 km: approximately 68.1%.
- Reminder-channel observed rates.
- Appointment-type observed rates.
- Risk-group observed rates: 69.98%, 55.70%, 38.55%.
- Limitations and KPI definitions.

Expected Data Science feedback

- Feature importance for previous no-show, lead time, distance and reminder variables.
- Model performance with and without candidate features.
- Precision, recall, F1 and/or other appropriate metrics for the no-show class.
- Calibration or probability quality if a risk score is produced.
- Whether the model confirms or challenges the descriptive patterns.
- Recommendations for features that should be retained, transformed or excluded.

8. Final Data Analytics Contribution to Project Management

Project Management should receive a concise decision table in the format: Finding → Evidence → Business implication → Recommended action → Validation status. This allows project decisions to be linked directly to analytical evidence.

9. Executive Decision Summary

The most actionable areas for further validation are previous no-show history, booking lead time, distance to clinic and risk-score segmentation. Reminder-channel differences should be investigated using adjusted or controlled analysis before operational conclusions are made. The final analytical objective is to convert these validated descriptive patterns into reliable predictive and operational decision support.